

A PROJECT REPORT ON

**CHATBOT-ENHANCED PLACEMENT DATA
MANAGEMENT AND SUMMARIZATION**

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FOR THE AWARD OF THE DEGREE

OF

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CERTIFICATE

This is to certify that the preliminary project report entitled

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ABSTRACT

This project is designed to improve the management and accessibility of placement data for students by utilizing automated data collection and sophisticated natural language processing techniques. Data will be automatically collected from seniors who have secured placements through a structured form. This information will then be summarized using a text summarization tool to extract essential insights, making it more comprehensible. Additionally, a chatbot, trained on the gathered data, will offer real-time support by responding to inquiries about various companies. Both the chatbot and the text summarizer are specifically developed to effectively address this use case.

Keywords: Automated data collection, Text summarization, Chatbot, Essential insights, Query response, Information management, Natural Language Processing (NLP), Large Language Model (LLM)

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01. INTRODUCTION

In today's digital age, the rapid growth of textual data poses considerable challenges for effective processing and utilization. This issue is especially relevant in educational environments, where students preparing for placements must sift through extensive information regarding company recruitment processes, experiences shared by seniors, and interview advice. Traditionally, managing this data has relied on manual efforts, which can be both inefficient and time-consuming. The increasing demand for more streamlined methods of processing such information has prompted the exploration of automated systems that can enhance the collection and accessibility of pertinent insights.

This project, therefore, seeks to improve the management and access of placement data by utilizing automated data collection and advanced natural language processing (NLP) techniques. One of the key components of this project is text summarization, which addresses the challenge of processing large volumes of unstructured textual data. While manual summarization can effectively capture main ideas, it requires considerable time and effort, making it impractical for large datasets. Automated text summarization (ATS) technologies can efficiently generate concise summaries that highlight key insights while maintaining the meaning and context of the original content.

This approach is particularly beneficial in the placement context, allowing students and administrators to quickly understand essential points from extensive textual data, such as feedback from seniors who have secured placements. By employing ATS, users can significantly reduce the time and effort needed to extract relevant insights without having to read through lengthy documents.

Additionally, this project incorporates a chatbot as an interactive interface to assist students by providing real-time responses to their inquiries about placement information. Chatbots, powered by NLP, are increasingly being adopted across various sectors, including education, due to their ability to simulate conversations and deliver personalized information instantly. Implementing such a system in a placement context enables students to engage with the data more intuitively.

The chatbot, trained on the information gathered from placed seniors, will be capable of answering a wide range of questions, offering a more accessible and user-friendly way for students to obtain placement-related information without the need to sift through numerous reports manually. The evolution of chatbot technology, from early conversational systems like ELIZA to modern virtual assistants such as Siri, Google Assistant, and IBM Watson, has made these systems more intelligent and adept at handling complex tasks.

Today's chatbots are platform-independent, requiring no installations and providing instant access across multiple devices, making them ideal for educational applications. These systems not only facilitate secure interactions but can also manage multiple conversations simultaneously, which is essential when addressing queries from numerous students about different companies.

In this project, the chatbot and text summarizer will collaborate to create a seamless experience for students by delivering essential insights from placement data in real time. This integration of ATS and chatbot technologies will streamline the processes of data collection and summarization while enhancing the overall accessibility and usability of placement information. By automating these tasks, students will be empowered to make more informed decisions with minimal effort, ultimately improving their preparation for recruitment and placement activities.

Through the incorporation of these advanced technologies, this project aims to transform the management of placement data, turning a traditionally labor-intensive task into an efficient and intelligent system. The chatbot's capability to provide real-time assistance, combined with the summarizer's ability to extract valuable insights from unstructured data, represents a significant advancement in how placement information can be processed and utilized by students, researchers, and administrators alike.

1.1) MOTIVATION

In the contemporary competitive campus placement environment, students preparing for recruitment often encounter challenges in gathering accurate and actionable data about various companies, their recruitment processes, and the preparation strategies of previously placed students. Understanding past trends in campus placements can significantly enhance the chances of success, but the absence of a centralized, streamlined system to collect and analyze such data creates a gap in students' preparation efforts. This project addresses that gap by developing an automated data collection and management platform that captures detailed information from students who have been successfully recruited.

The core of this project involves gathering data related to students' recruitment experiences, company profiles, and the preparation strategies that led to their success. To ensure ease of use, the system is designed for administrators to enter minimal data, while the platform automatically processes, organizes, and updates the information. One of the key features of the system is its text summarization capability, which aggregates the reports of all students placed in the same company during a particular year. This provides

a clear summary of that company's recruitment trends, helping students identify patterns and key insights that can impact their own preparation strategies.

In addition to data collection and summarization, the system also includes a chatbot feature powered by the entire dataset. This chatbot is capable of handling complex user queries, allowing students to easily access specific information regarding placement processes, company expectations, and preparation tips. The chatbot enhances the accessibility of the platform, enabling users to quickly obtain tailored responses based on historical data.

By providing a comprehensive view of past recruitment experiences and trends, this project offers significant value to students in their placement preparation journey. The insights generated by the platform can help them make better-informed decisions, refine their preparation strategies, and understand company-specific recruitment dynamics. Ultimately, this system serves as a robust tool for placement preparation, bridging the information gap and empowering students to approach their recruitment with confidence and clarity.

1.2) PROBLEM DEFINITION

In the campus placement process, students often encounter challenges in accessing accurate, comprehensive, and actionable information about the recruitment procedures of various companies. The lack of a centralized system for gathering and analyzing the experiences of previously placed students creates a significant information gap. Consequently, students may find it difficult to prepare effectively for placements, missing out on valuable insights regarding company-specific trends, recruitment patterns, and successful preparation strategies.

Additionally, there is no streamlined approach for administrators to efficiently collect, organize, and update placement-related data. The manual collection of such information can be time-consuming and prone to errors, further exacerbating the lack of accessible, reliable information for students.

This project aims to tackle these challenges by developing an automated data collection and management platform that captures detailed information from students who have successfully navigated the recruitment process. The system will summarize this data and provide key insights through a text summarization feature, presenting clear trends for each company's recruitment process. Furthermore, the integration of a chatbot will enable students to easily query the dataset and access specific information, thereby enhancing the overall placement preparation process.

The absence of a centralized, efficient data management system for campus placements, along with the difficulties students face in accessing reliable information, forms the core problem that this project seeks to address.

02. LITERATURE SURVEY

Work	ML Algorithm	Features
A Generative Adversarial Network-Based Extractive Text Summarization Using Transductive and Reinforcement Learning	RL algorithm	Improving local sentences
A Review on Optimization-Based Automatic Text Summarization Approach	Artificial Bee Colony (ABC) algorithm	prioritizes factual accuracy
Advanced Text Summarization Model Incorporating NLP Techniques and Feature-Based Scoring	fine-tuning	Named Entity Counts, Noun Counts
Next-Generation Text Summarization: A T5-LSTM FusionNet Hybrid Approach for Psychological Data	K-Means	sequential learning
Explainable AI Chatbots towards XAI ChatGPT	Large Language Models (LLMs), Hybrid (e.g. GANs, Knowledge Graphs)	Emphasis on explainability, transparency, human trust, counterfactual reasoning, sector-specific features
Artificial Intelligence Based Chatbot for Placement Activity at College Using DialogFlow	NLP via DialogFlow (Intent + Fulfillment logic)	Uses predefined intents, training phrases, fulfillment API; improves placement-related info access
A Review on Building Customized Chatbots for Document Summarization and Question Answering Using OpenAI, LangChain and Streamlit	OpenAI GPT models with LangChain Framework	Uses LLMs for summarization and Q&A; integrates NLP pipeline, UI via Streamlit
A Complete Survey on LLM based Chatbots	Deep Learning (Transformers-based)	Context-aware summarization, uses domain-specific embeddings
Building Customized Chatbots for Document Summarization and Q&A using LLMs with OpenAI, LangChain, and Streamlit	GPT (LLMs) + LangChain	End-to-end document summarization, question answering, Streamlit UI

03. SOFTWARE REQUIREMENTS SPECIFICATION

The software requirements specification outlines the essential software components, tools, and frameworks required for the development and successful deployment of the proposed system. It ensures the compatibility, efficiency, and scalability of the system, focusing on both development and user interaction aspects. The system is based on a full-stack architecture, incorporating machine learning and natural language processing capabilities.

1. Operating System

The system can be developed and executed on any modern operating system that supports Python and Django. Preferably, the system should run on:

- Windows 10/11
- Ubuntu 20.04+
- macOS (latest versions)

These operating systems provide the necessary compatibility for both backend and machine learning development environments.

2. Programming Language

- Python 3.8 or higher is used as the primary programming language due to its simplicity, robust support for machine learning, and availability of extensive libraries.
- Python is suitable for both the backend development using Django and for implementing machine learning models for text summarization and chatbot functionalities.

3. Backend Framework

- Django Framework is used for server-side development.
- Django provides built-in functionalities for handling database models, authentication, routing, and REST API creation, which simplifies development and enhances security and scalability.

4. Frontend Technologies

- HTML5 and CSS3 are used for designing the structure and styling of the web interface.
- Bootstrap 4/5 is used to create a responsive and user-friendly frontend interface.

5. Machine Learning and NLP Libraries

The following libraries are used for developing text summarization and chatbot modules:

- NumPy and Pandas: For data preprocessing and numerical operations.
- Scikit-learn: For machine learning model training and evaluation.
- TensorFlow and PyTorch: For building and training deep learning models.
- Transformers (from Hugging Face): For using pre-trained large language models (LLMs) for summarization and chatbot tasks.

6. Evaluation and Testing Tools

- Postman: For testing REST APIs and verifying endpoints of the backend system.
- Scikit-learn Evaluation Metrics: Tools like BLEU, ROUGE, and F1-score are used to evaluate the performance of NLP models.
- Django's Testing Framework: Used to test backend logic, views, and models.

7. Database

- SQLite is used during the development phase for its simplicity and lightweight nature.
- PostgreSQL or MySQL may be used in production for better performance and data handling capabilities.

8. Integrated Development Environment (IDE)

- Visual Studio Code, PyCharm, or Jupyter Notebook are used for code writing, debugging, and model experimentation.

9. Version Control

- Git is used for version control, enabling collaborative development and tracking of changes.
- GitHub or GitLab is used as the remote repository for source code management.

10. Web Browser

- The application is compatible with all modern browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge, ensuring a consistent user experience across platforms.

11. Optional Tools for Deployment (if applicable)

- Docker: For containerizing the application for deployment across different environments.

- Streamlit/Flask: For demo interfaces or showcasing ML functionalities in a user-friendly manner.

3.1) INTRODUCTION

3.1.1) Project Scope

The project aims to create an automated data collection and management platform specifically for campus placement preparation. This platform streamlines the collection, summarization, and analysis of recruitment-related data from students who have previously secured placements, thereby providing actionable insights for those preparing for upcoming placement processes. Key features of the platform include text summarization of placement experiences, a chatbot for querying the dataset, and a user-friendly interface designed for both administrators and students. The system is intended to enhance placement preparation by delivering reliable, data-driven insights into organization-specific recruitment trends.

3.1.2) Use Cases and Characteristics

The primary use cases of the system include:

- **Data Collection:** Administrators input recruitment-related data from students who have successfully completed the placement process.
- **Data Summarization:** The system automatically processes and summarizes data for each organization, highlighting key recruitment patterns.
- **Student Interaction through Chatbot:** Students can interact with the chatbot to access specific information regarding companies, placement processes, and preparation techniques.

Characteristics of the System:

- **Automated Data Processing:** The platform processes data with minimal manual input required.
- **Real-Time Query Handling:** The chatbot provides real-time responses to student queries.
- **Scalability:** The system is designed to accommodate an increasing volume of data entries from both students and organizations.

3.1.3) Assumptions and Dependencies

Assumptions:

- Students are open to sharing detailed accounts of their placement experiences.
- Recruitment trends among employers can be identified through historical data.
- Administrators are tasked with ensuring the accuracy of data entered into the system.

Dependencies:

- The system's functionality depends on accurate data input from administrators.
- The effectiveness of text summarization and chatbot features relies on pre-trained machine learning models.

3.2) Functional Requirements

3.2.1) System Feature 1 (Practical Requirement)

The primary feature of the system focuses on the automated gathering and management of student placement information. Administrators will be provided with an intuitive interface that allows for easy entry and updating of recruitment-related data. To uphold accuracy and integrity during data entry, the system will implement data validation checks. It will include dropdown menus for standardized responses and notifications for any incomplete or erroneous submissions. This functionality is essential for obtaining thorough and dependable information regarding previous placement experiences.

3.2.2) System Feature 2 (Functional Requirement)

The second feature of the system includes text summarization and chatbot capabilities. The platform will leverage Natural Language Processing (NLP) and Large Language Model (LLM) techniques to examine student placement reports and produce brief summaries that emphasize key recruitment trends for each organization. Furthermore, the chatbot will be equipped to address real-time inquiries from students, enabling them to obtain specific details about companies, placement procedures, and preparation strategies. It will also employ machine learning to improve its responses over time based on user interactions.

3.3) External Interface Requirements

3.3.1) User Interfaces

The system is designed to provide a clean, accessible, and user-friendly interface for students and administrators to interact with placement-related data.

- Technologies Used: HTML5, CSS3, Bootstrap 4/5
- Design Goal: Desktop-optimized interface for intuitive navigation.
- Key Features:
 - Home page with chatbot interface
 - Upload section for senior experience documents
 - Text summarization results display
 - Query input and chatbot output section
- Responsiveness: Designed primarily for desktop and laptop screens.

3.3.2) Hardware Interfaces

No specialized hardware interfaces are required. The system is intended to be run on standard personal computers or laptops with basic configuration.

- Input Devices: Keyboard, Mouse
- Output Devices: Display screen (minimum resolution 1366x768)
- Optional: Microphone (for future voice-enabled chatbot integration)

3.3.3) Software Interfaces

The software interacts with various components and frameworks to deliver its functionality.

- Backend Framework: Django (Python-based)
- Database: SQLite (for development and lightweight deployment)
- Frontend Technologies: HTML, CSS, Bootstrap
- Libraries and Frameworks:
 - Machine Learning: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch
 - NLP: Hugging Face Transformers
 - API Testing: Postman
- Operating Systems Supported: Windows 10/11, Ubuntu 20.04+, macOS

3.3.4 Communication Interfaces

The communication within the system components and between client-server follows standard protocols and techniques.

- Client-Server Communication: HTTP/HTTPS using REST API endpoints
- Data Format for Transmission: JSON (JavaScript Object Notation)
- Frontend to Backend: AJAX calls or form submissions
- External Communication: None currently implemented (future versions may integrate third-party APIs)

3.4) Non-Functional Requirements

3.4.1) Performance Requirements

The system is required to uphold high performance standards, ensuring it can accommodate a large number of simultaneous users without a decline in response times. Queries and interactions with the chatbot should be processed within a matter of seconds to improve the user experience. Load testing will be performed to verify that the system can manage peak usage during placement seasons effectively.

3.4.2) Safety Requirements

Safety measures will be established to safeguard the integrity of the data stored in the system. This will involve regular database backups to prevent data loss and mechanisms to ensure that data remains uncorrupted during processing. Additionally,

the platform will include error handling procedures to address and log any unexpected issues that may arise.

3.4.3) Security Requirements

The system will incorporate strong security measures to safeguard sensitive information. Role-Based Access Control (RBAC) will be employed to guarantee that only authorized users have the ability to access or modify sensitive data. Furthermore, the system will utilize OAuth or JWT for secure user authentication, ensuring that only verified users can gain access to the platform.

3.5) System Requirements

3.5.1) Database Requirements

The database will be structured to effectively manage and store substantial amounts of recruitment data. It must be capable of handling complex queries for reporting and analysis while being scalable to support a growing number of records over time. The database schema will be optimized for performance, facilitating rapid retrieval of information.

3.5.2) Software Requirements (Platform Choice)

The project utilizes Django as the backend framework due to its scalability, security, and compatibility with Python-based machine learning models. For the frontend, HTML, CSS, and Bootstrap have been selected to create a clean and structured interface that is desktop-friendly. SQLite serves as the lightweight database that integrates seamlessly with Django for efficient data storage. Python is the primary programming language, chosen for its robust support for Natural Language Processing (NLP) and machine learning, utilizing libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and PyTorch. Postman is employed for API testing, while Scikit-learn's evaluation metrics are used to assess the performance of machine learning models. This technology stack ensures smooth integration between web and machine learning components while maintaining efficient and manageable development processes.

3.5.3) Hardware Requirements

The hardware requirements for deploying the system will consist of standard server configurations that can host both the application and the database. These servers should possess adequate processing power, memory, and storage to effectively handle user requests and manage data. Local development can be carried out on standard personal computers, ensuring accessibility for developers.

04. PROJECT PLAN

4.1) ESTIMATE

The project estimate includes a comprehensive analysis of the resources, time, and budget necessary to complete the system's development. Key components of the estimate are as follows:

- **Resource Allocation:** Identification of key team members, such as developers, designers, and testers. Each role will be assigned according to their expertise and the specific needs of the project.
- **Timeframe:** A detailed breakdown of the project timeline, highlighting key milestones such as requirement gathering, design, development, testing, and deployment phases. Each phase will be assigned a specific timeframe based on its complexity and the availability of the team.
- **Budget:** An overview of anticipated costs, including personnel expenses, software licenses, hardware acquisitions, and any third-party services needed (e.g., cloud hosting, APIs). The budget will also incorporate a contingency fund to cover any unforeseen expenses.

4.2) RISK MANAGEMENT

The risk management strategy is designed to systematically identify, evaluate, and address potential risks that may affect the project. The key components of this strategy include:

- **Risk Identification:** Conducting a comprehensive analysis to uncover potential risks, such as technical difficulties, availability of resources, evolving requirements, and budget limitations.
- **Risk Assessment:** Assessing the probability and potential impact of each identified risk, categorizing them into high, medium, or low levels. This evaluation will inform the prioritization of risk mitigation efforts.
- **Mitigation Strategies:** Formulating contingency plans for high-impact risks, which may involve reallocating resources, exploring alternative technologies, or modifying project timelines. Continuous monitoring of risks throughout the project lifecycle will facilitate prompt responses to any emerging challenges.

4.3) PROJECT SCHEDULE

4.3.1) Project Task Set

The project schedule delineates the timeline for all project activities, specifying the sequence of tasks and their interdependencies. The key components of this schedule include:

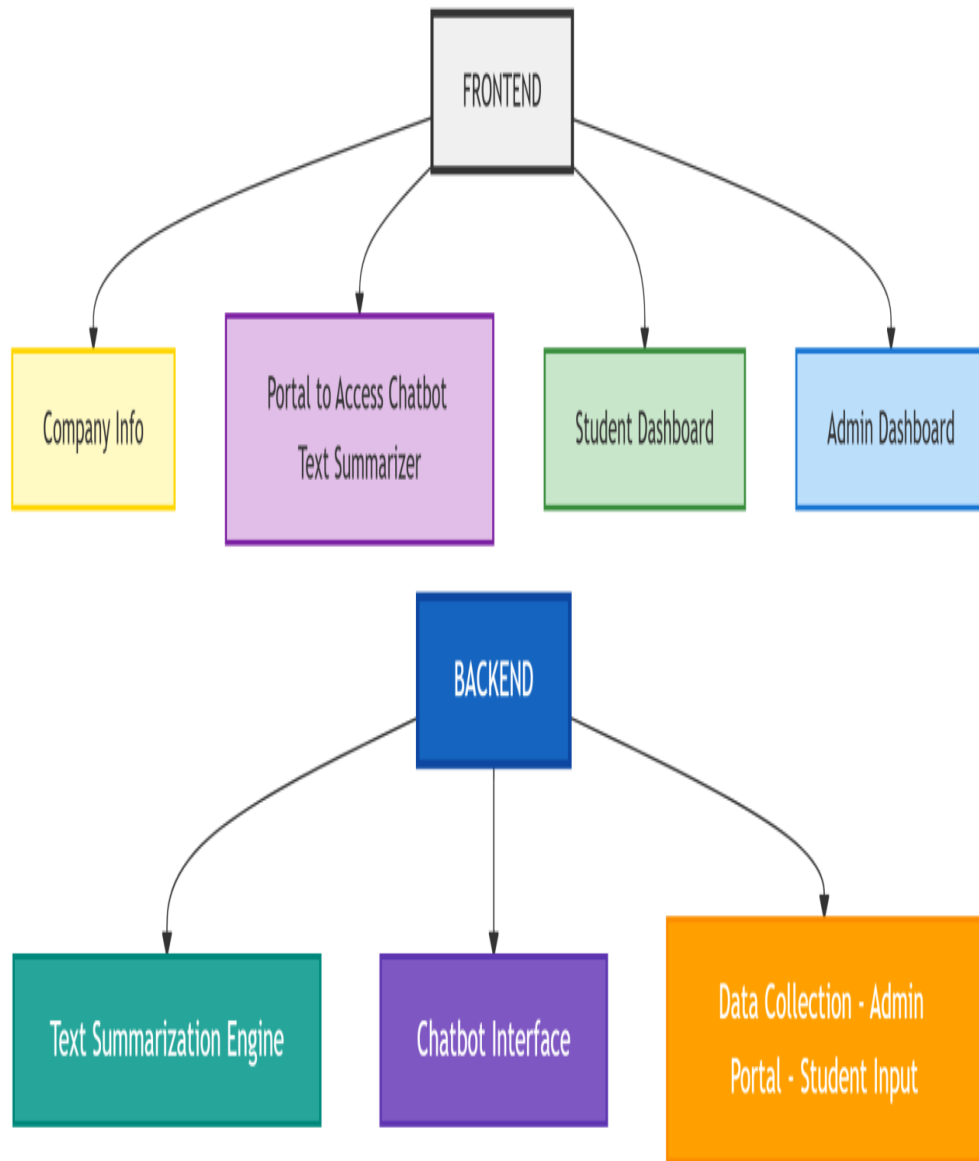
- **Gantt Chart:** A graphical representation of the project timeline that displays the start and end dates for each task, along with milestones and critical paths. The

Gantt chart will act as a communication tool to keep stakeholders updated on project progress.

- **Milestones:** Identification of significant milestones, such as the completion of requirement analysis, design approval, the conclusion of development, and successful testing phases. These milestones will aid in tracking progress and ensuring timely project delivery.
- **Task Breakdown:** A comprehensive breakdown of all tasks necessary to achieve each milestone, detailing task owners, expected durations, and dependencies. This breakdown will support effective project management and resource allocation.

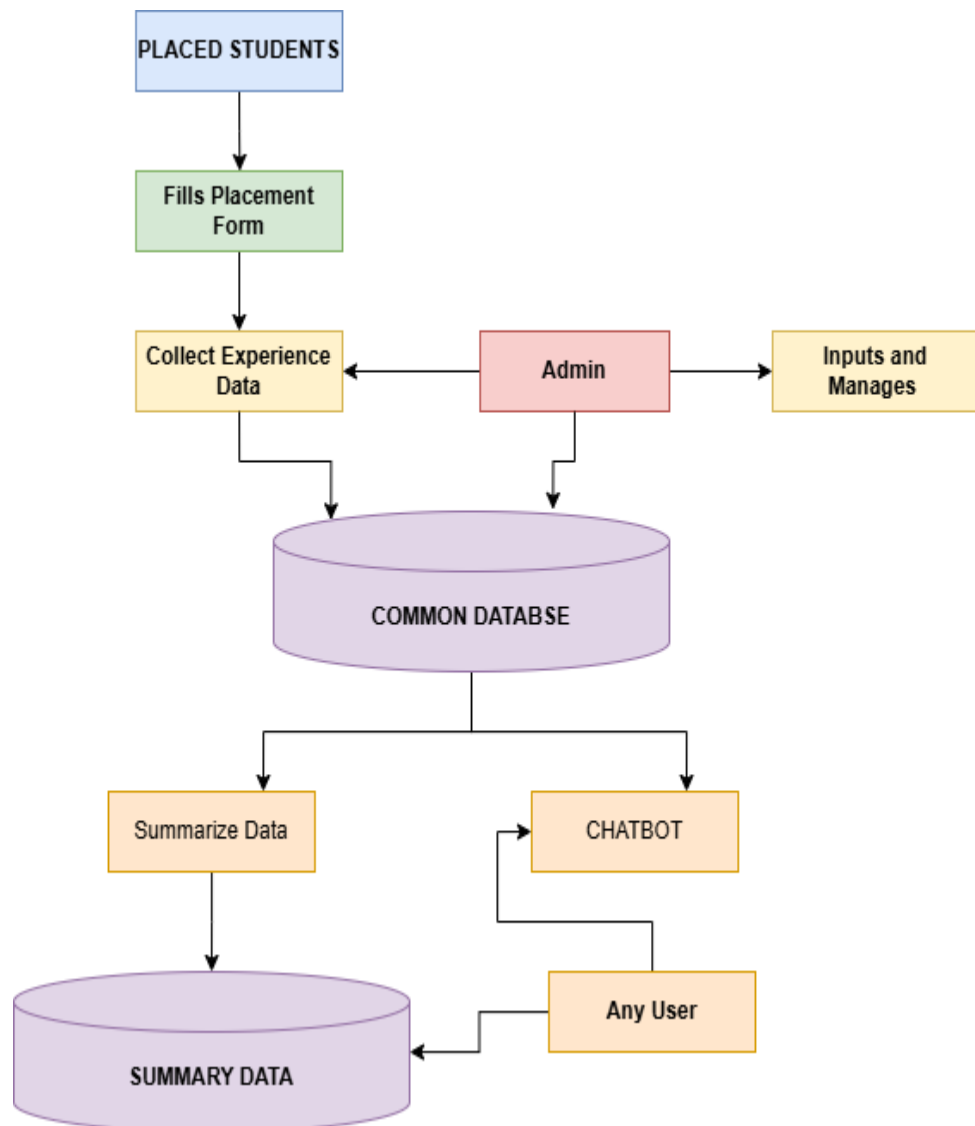
05.SYSTEM DESIGN

5.1) SYSTEM ARCHITECTURE

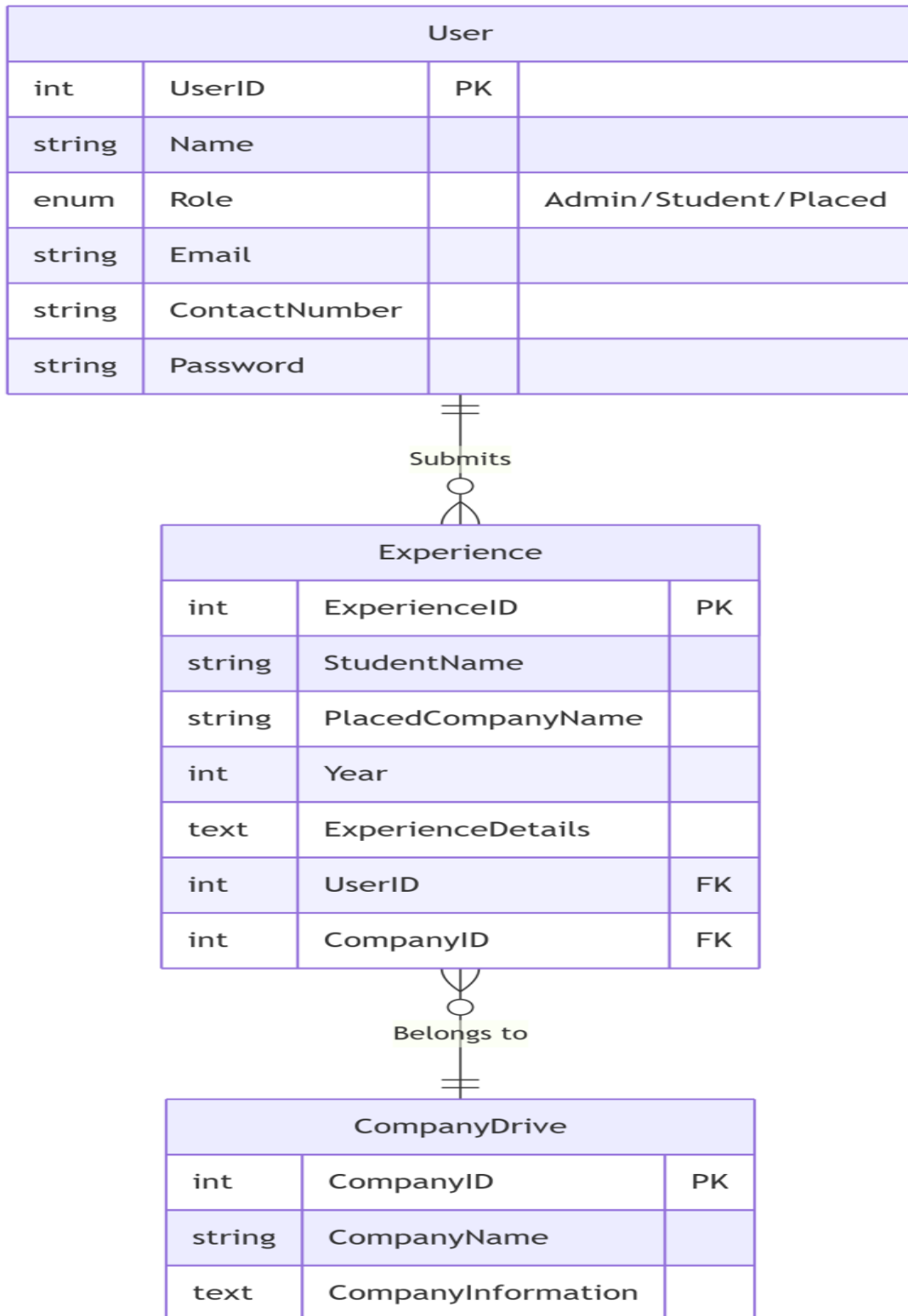


This system architecture is designed to manage recruitment data, provide text Summarization and chatbot, and offer insights for students and admins.

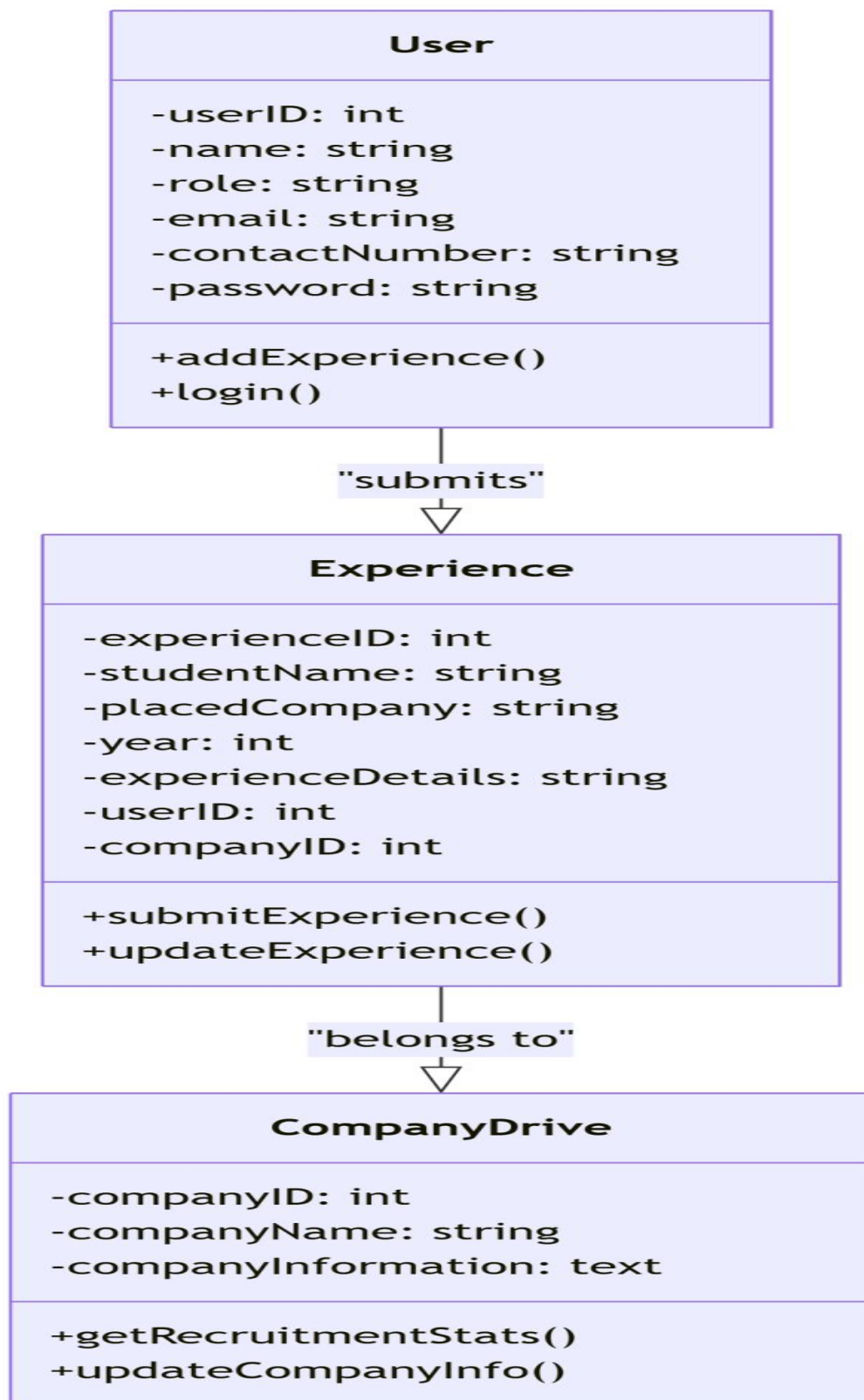
5.2) DATA FLOW DIAGRAMS



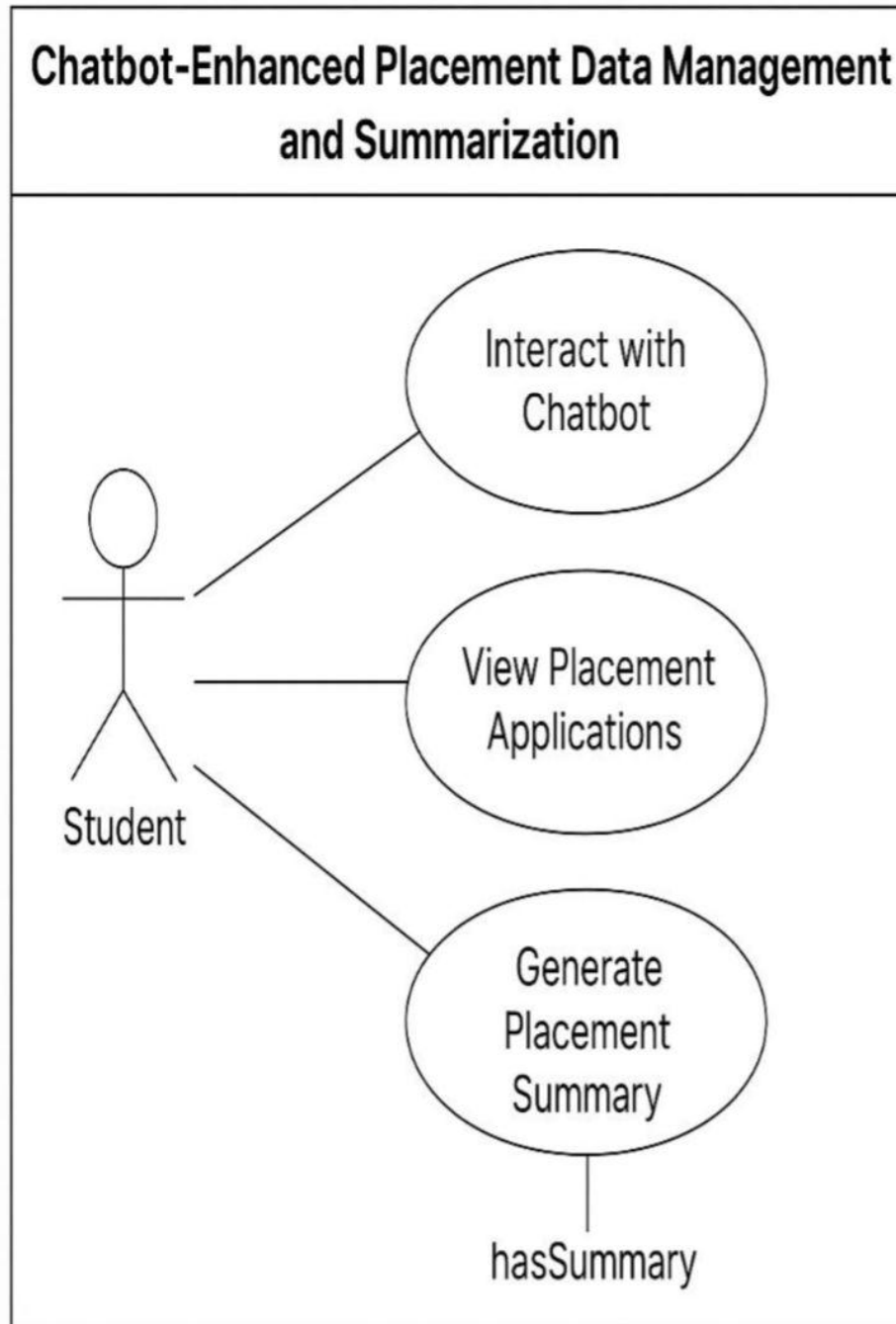
5.3) ENTITY RELATIONSHIP DIAGRAMS



5.4) UML DIAGRAM



5.5) USE CASE DIAGRAM:



06. OTHER SPECIFICATIONS

6.1) ADVANTAGES

The placement management system provides several key advantages:

- **Streamlined Processes:** The system automates various recruitment-related tasks, significantly reducing the time and effort needed for data entry, management, and reporting. This enhancement leads to greater efficiency and productivity for both administrators and students.
- **Enhanced Data Accuracy:** By incorporating validation checks and standardized inputs, the system ensures consistent and accurate data entry. This reduces errors in placement records and enhances the reliability of the information provided.
- **User -Friendly Interface:** The platform's intuitive design allows both administrators and students to easily navigate and access the information they require, promoting a positive user experience.
- **Real-Time Access to Information:** Features such as chatbots and dynamic dashboards provide users with real-time insights into placements, trends, and recruitment processes, facilitating informed decision-making.
- **Scalability:** The system is built to grow alongside the evolving needs of educational institutions, capable of handling an increasing volume of data and users over time.

6.2) LIMITATIONS

While the placement management system offers numerous benefits, it also encounters certain limitations:

- **Dependence on Technology:** The system's effectiveness hinges on the availability and reliability of technology, including internet access and server uptime. Any technical difficulties can impede user access and functionality.
- **Initial Development Costs:** The development and implementation of the system may incur substantial upfront costs associated with software development, hardware acquisition, and user training.
- **User Adoption:** There may be resistance to adopting the new system among users who are accustomed to traditional methods of managing placement data. Comprehensive training and support will be necessary to ensure a smooth transition.
- **Data Security Concerns:** Given that the system handles sensitive information, there is an inherent risk of data breaches or unauthorized access. Continuous security measures will be crucial to safeguard user data.

6.3) APPLICATIONS

The placement management system can be utilized in a variety of contexts:

- Educational Institutions: Colleges and universities can implement this system to manage student placements, track recruitment activities, and enhance communication between students and prospective employers.
- Recruitment Agencies: Agencies can leverage the platform to oversee candidate placements, streamline their operational processes, and deliver comprehensive reports on recruitment trends to their clients.
- Corporate Training Programs: Organizations that offer training programs can use the system to monitor participant placements and evaluate the effectiveness of their training in facilitating job placements.
- Career Development Centers: Career services can utilize the system to provide resources and support to students seeking employment, thereby improving their placement opportunities.
- Data Analysis and Research: The data collected through the system can be analyzed for research purposes, yielding insights into industry trends, student preferences, and the effectiveness of various placement strategies.

07.CONCLUSIONS AND FUTURE WORK

CONCLUSION

This project addresses the growing complexity of managing and accessing placement data for students by integrating AI technologies such as text summarization and chatbots. The automated collection and processing of data not only streamline information flow but also enhance accessibility and utility for students preparing for placements. By tackling issues like the limited use of AI in placement systems, difficulties in processing unstructured data, and the absence of real-time, personalized assistance, this solution offers a contemporary approach to resolving inefficiencies in placement preparation.

The application of advanced natural language processing techniques ensures that data is summarized and presented in a meaningful manner, enabling students to make informed decisions based on concise, actionable insights. Furthermore, the chatbot provides an interactive platform that allows students to quickly and easily find relevant information, thereby improving their overall placement experience.

This system has the potential to significantly transform how placement data is managed and utilized within educational institutions, laying the groundwork for further AI-driven innovations in student support systems.

FUTURE WORK

While the current system provides significant improvements, there are several areas that can be further enhanced in future iterations:

- **Integration with External Job Portals:** Future enhancements could involve integrating the system with external job portals to offer students a wider array of job opportunities.
- **Advanced Analytics and Reporting:** Incorporating more advanced data analytics features, such as predictive analytics and comprehensive trend analysis, could enable administrators to gain deeper insights into recruitment patterns and prepare more effectively.
- **Mobile Application Development:** Developing a mobile application version of the platform could improve accessibility, allowing students and administrators to access placement information and updates while on the go.
- **Artificial Intelligence (AI) Integration:** The addition of AI-driven chatbots for more sophisticated query handling, or the use of machine learning algorithms to recommend personalized job opportunities to students, could further enhance the user experience.

- **Multi-Institutional Support:** Expanding the system to support multiple institutions could facilitate collaborative placement activities and inter-college recruitment drives, creating more opportunities for students.

APPENDIX A: PROBLEM STATEMENT FEASIBILITY ASSESSMENT

The primary issue addressed by this project was the inefficiency and inaccuracy associated with traditional placement management methods. The system evaluated the feasibility of automating the recruitment workflow across several key areas:

- **Technical Feasibility:** The proposed system was deemed technically feasible, as the necessary technologies, including databases, web platforms, and cloud services, are readily accessible. The development team successfully implemented core functionalities such as data management, reporting, and real-time updates, confirming the technical viability of the solution.
- **Operational Feasibility:** The system integrates well with the daily operations of academic institutions, streamlining workflows and enhancing productivity. Both administrators and students found the system intuitive and beneficial, demonstrating its operational effectiveness.
- **Economic Feasibility:** Although initial development costs were substantial, the long-term savings in terms of time, reduced manual labor, and fewer errors render the system economically viable. The return on investment (ROI) is anticipated to increase as the system scales and becomes a vital tool for managing placements.
- **Legal and Security Feasibility:** Data security and privacy considerations were integral to the system's design, with appropriate measures implemented to comply with data protection regulations. This ensures the system's feasibility in managing sensitive student and recruitment data.

APPENDIX B: Plagiarism Report

APPENDIX C: Survey Paper